Sensitivity analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete sensitivity range and evidence record.

Keep controls, settings, outcomes, missing labels and selection disclosures together.

Keep one habit: Recall → W1–W3/W5 → independent pass-count comparison → research project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

Ablation studies

My remembered idea, example and correction

Statistical tests and assumptions

My remembered idea, example and correction

Sampling variability

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Sensitivity analysis varies plausible choices across a declared range and preserves all results. It describes dependence on choices; choosing the best result becomes tuning and needs separately reserved evaluation for a fresh performance claim. Main retrieval depths 1, 3, 5 have stipulated recall 0.4, 0.7, 0.8 and precision 0.9, 0.7, 0.5.

  Recall rises and precision falls across this tested range; no universal best follows without an objective. Raw labels, counts and averaging convention are not supplied. Separate picture scores 0.4, 0.6, 0.8 stay fixed. Positive means score at least cutoff. Cutoffs 0.5 and 0.6 give negative/positive/positive, while 0.7 gives negative/negative/positive. Equality is included.

  Positive counts are two, two and one respectively, not correctness counts. No reference labels are supplied. Hypothetical alternative labels can favor either rule and must not be recorded as observed truth. Keep data roles and all tested settings visible; no actual model study, significance result or guarantee outside this range is supplied.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- Sensitivity analysis varies plausible choices across a declared range and preserves all results. It describes dependence on choices; choosing the best result becomes tuning and needs separately reserved evaluation for a fresh performance claim. Main retrieval depths 1, 3, 5 have stipulated recall 0.4, 0.7, 0.8 and precision 0.9, 0.7, 0.5.

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- Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

Retrieved items k	Supplied recall	Supplied precision	Pattern in the tested range
1	0.4	0.9	Lowest recall; highest precision
3	0.7	0.7	Intermediate summaries
5	0.8	0.5	Highest recall; lowest precision
From k = 1 to k = 5	Increase 0.4, or 40 percentage points	Decrease 0.4, or 40 percentage points	Tradeoff; no universal best setting
What remains unknown	Raw labels/counts	Averaging convention	No invented per-query count or significance

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Sensitivity analysis varies plausible choices across a declared range and preserves all results. It describes dependence on choices; choosing the best result becomes tuning and needs separately reserved evaluation for a fresh performance claim. Main retrieval depths 1, 3, 5 have stipulated recall 0.4, 0.7, 0.8 and precision 0.9, 0.7, 0.5.

  Recall rises and precision falls across this tested range; no universal best follows without an objective. Raw labels, counts and averaging convention are not supplied. Separate picture scores 0.4, 0.6, 0.8 stay fixed. Positive means score at least cutoff. Cutoffs 0.5 and 0.6 give negative/positive/positive, while 0.7 gives negative/negative/positive. Equality is included.

  Positive counts are two, two and one respectively, not correctness counts. No reference labels are supplied. Hypothetical alternative labels can favor either rule and must not be recorded as observed truth. Keep data roles and all tested settings visible; no actual model study, significance result or guarantee outside this range is supplied.

- A report omits this qualification: A stable tested range does not establish stability under every parameter or dataset change. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Sensitivity analysis varies plausible choices across a declared range and preserves all results. It describes dependence on choices; choosing the best result becomes tuning and needs separately reserved evaluation for a fresh performance claim. Main retrieval depths 1, 3, 5 have stipulated recall 0.4, 0.7, 0.8 and precision 0.9, 0.7, 0.5.

  Recall rises and precision falls across this tested range; no universal best follows without an objective. Raw labels, counts and averaging convention are not supplied. Separate picture scores 0.4, 0.6, 0.8 stay fixed. Positive means score at least cutoff. Cutoffs 0.5 and 0.6 give negative/positive/positive, while 0.7 gives negative/negative/positive. Equality is included.

  Positive counts are two, two and one respectively, not correctness counts. No reference labels are supplied. Hypothetical alternative labels can favor either rule and must not be recorded as observed truth. Keep data roles and all tested settings visible; no actual model study, significance result or guarantee outside this range is supplied.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Thresholds .4, .5, .6 give passes 8, 9, 7 out of 10. Report range and a selection limit.
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **sensitivity analysis evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **sensitivity analysis**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Sensitivity analysis**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Sensitivity analysis**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Thresholds .4, .5, .6 give passes 8, 9, 7 out of 10. Report range and a selection limit.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

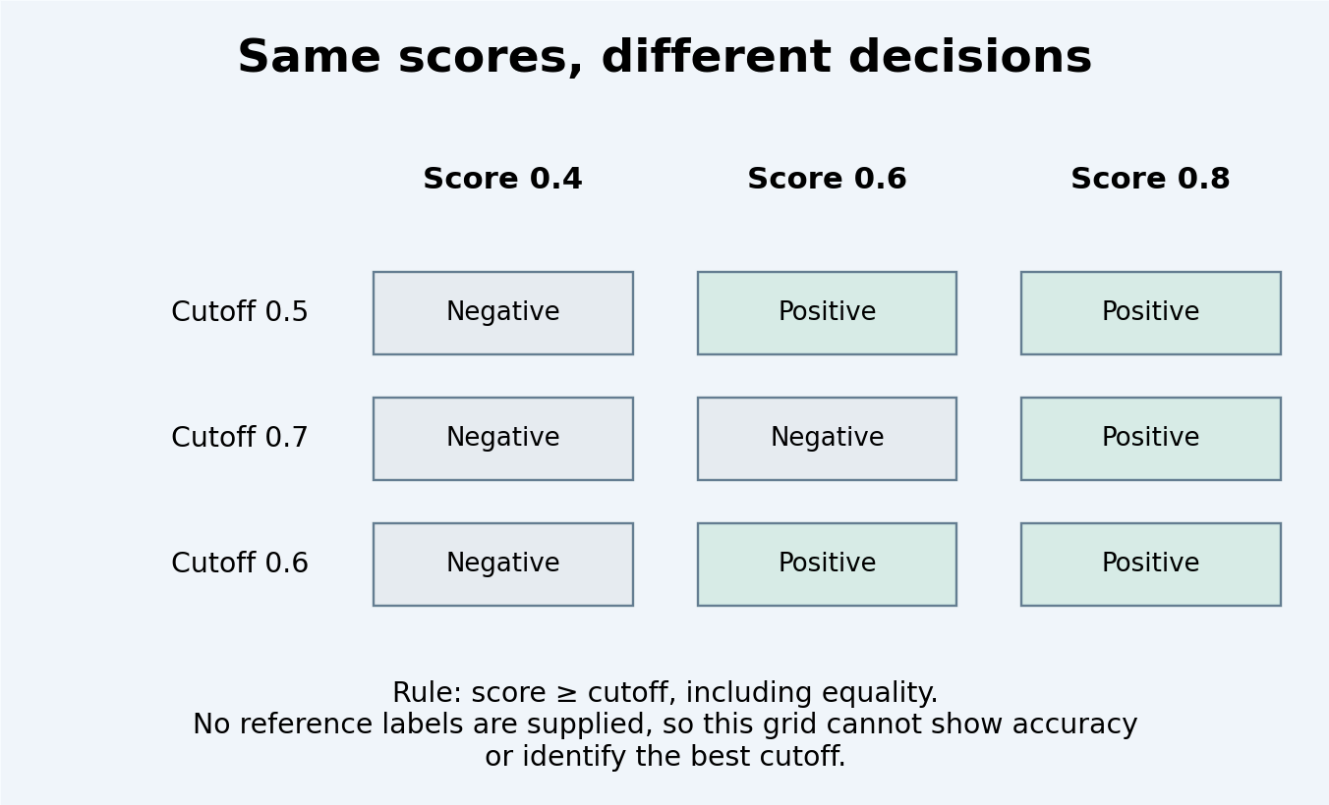
- Scores are 0.4,0.6,0.8. Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

- Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6. Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/17

Score decisions under inclusive cutoffs

	Score 0.4	Score 0.6	Score 0.8
Cutoff 0.5	No	Yes	Yes
Cutoff 0.7	No	No	Yes
Cutoff 0.6	No	Yes	Yes

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/17

Sensitivity analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Hold scores fixed

Keep the same three supplied scores.



2. Apply cutoff 0.5

0.6 and 0.8 meet the boundary.



3. Apply cutoff 0.7

Only 0.8 meets the new boundary.

CHANGE ONE THING

Set the cutoff to exactly 0.6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.
- Which labelled step matches this detail? Only 0.8 meets the new boundary.
- 1. Hold scores fixed
- 2. Apply cutoff 0.7
- 3. Apply cutoff 0.5
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply cutoff 0.7 Only 0.8 meets the new boundary. **Sensitivity analysis** checks how conclusions change under plausible choices.

Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

Changed-case question and optional hint

- Changed-case practice: Set the cutoff to exactly 0.6. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. 0.6 and 0.8 are positive because the rule includes equality.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: 0.6 and 0.8 are positive because the rule includes equality. The declared at-least boundary includes the threshold value.

Repair and retain the record

Review Ablation studies (ai-7/16). Goal: Explain the **mechanism** and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Review Sampling variability (ai-7/11). Goal: Explain the **mechanism** and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | |
> |---|
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

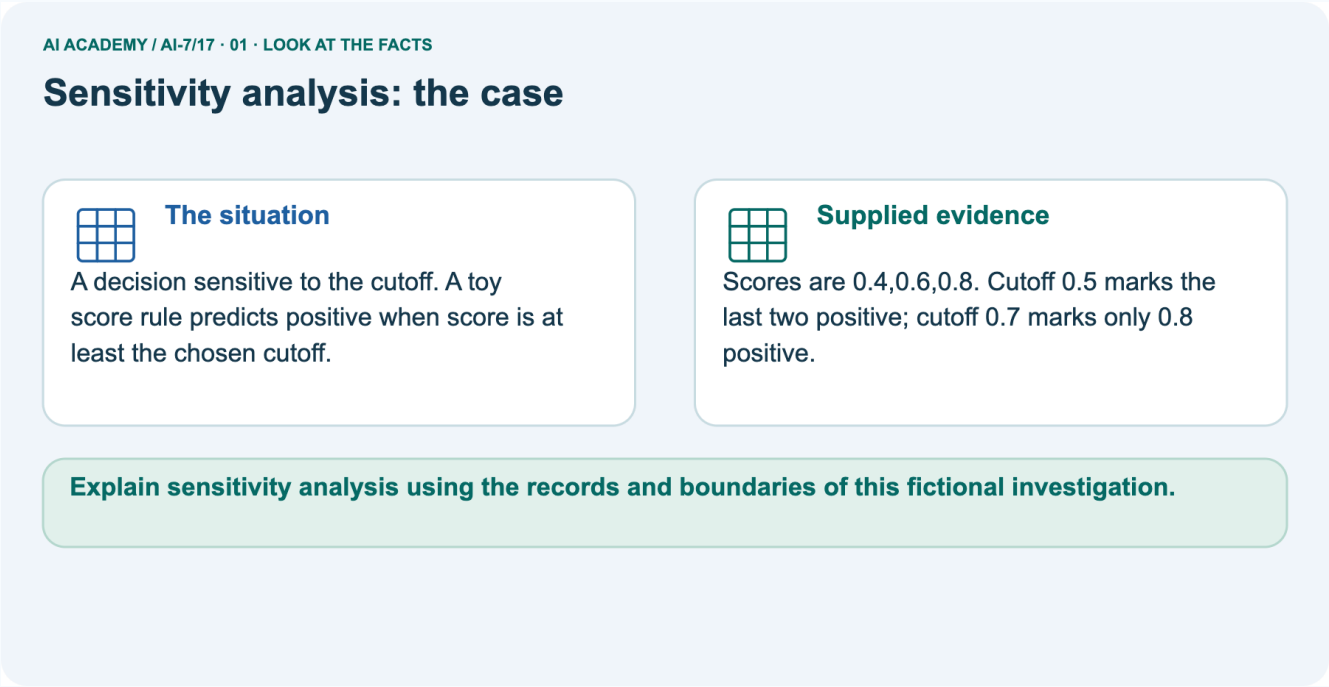
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Sensitivity analysis: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

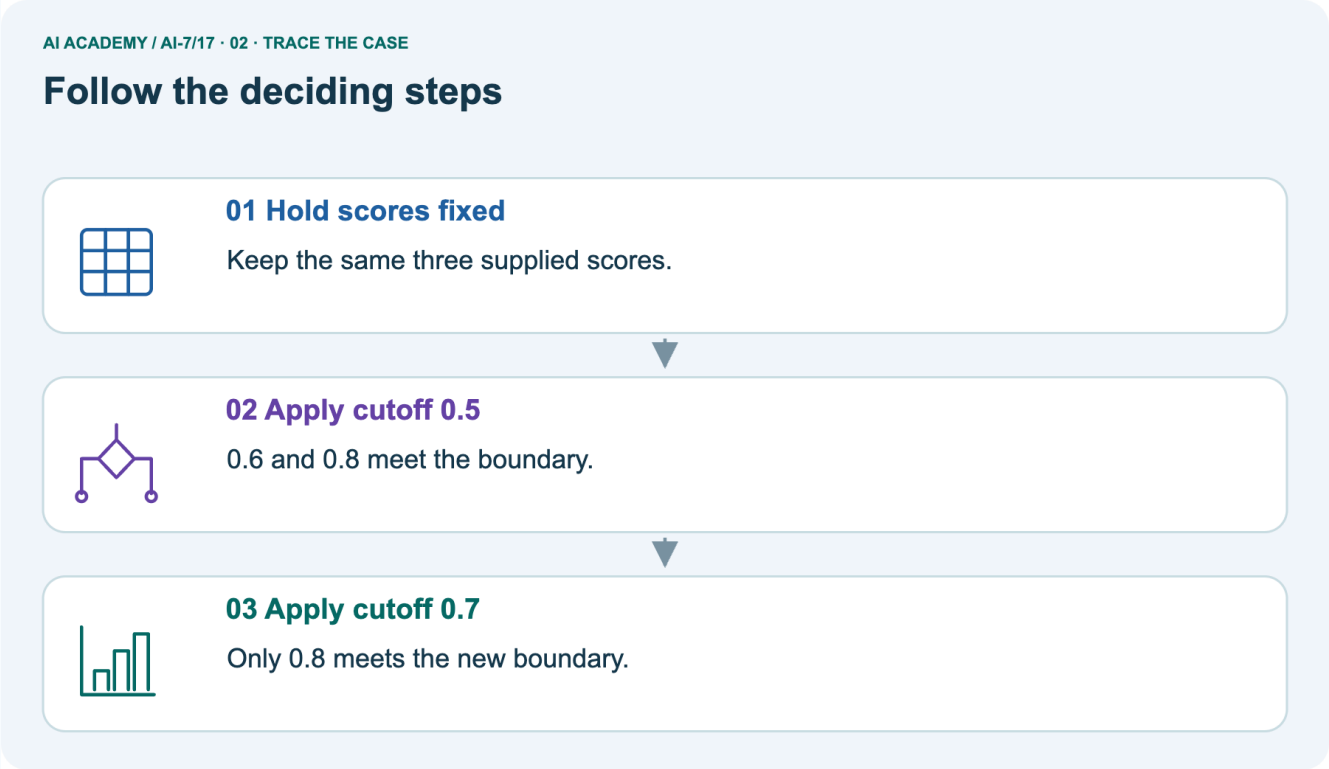
A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff. Supplied facts: Scores are 0.4,0.6,0.8.

Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



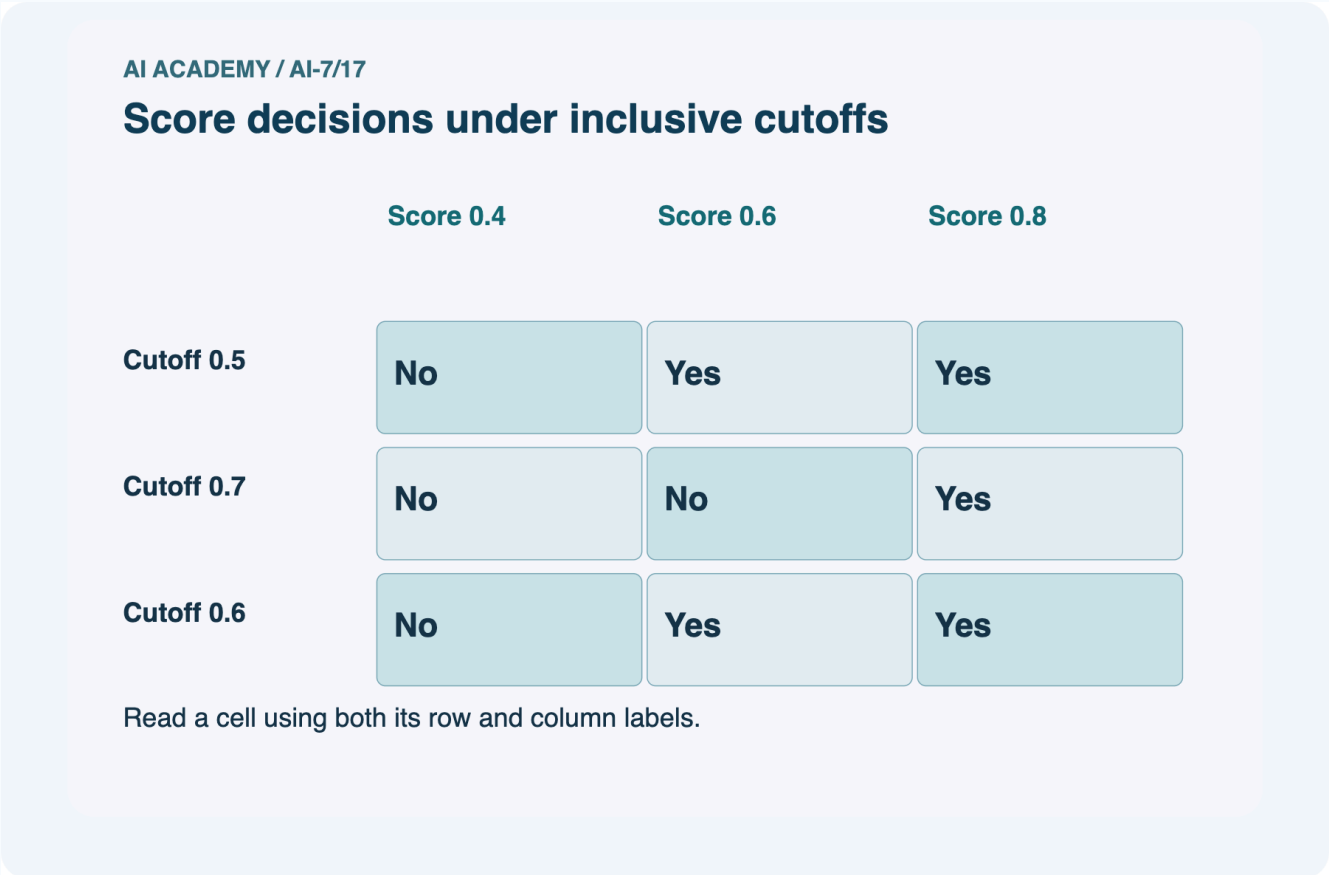
Follow the labelled picture with the matching explanation. Open the vector diagram

Hold scores fixed: Keep the same three supplied scores. → Apply cutoff 0.5: 0.6 and 0.8 meet the boundary. → Apply cutoff 0.7: Only 0.8 meets the new boundary.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Score decisions under inclusive cutoffs



Follow the labelled picture with the matching explanation. Open the vector diagram

Score decisions under inclusive cutoffs. Cutoff 0.5: Score 0.4 = No, Score 0.6 = Yes, Score 0.8 = Yes. Cutoff 0.7: Score 0.4 = No, Score 0.6 = No, Score 0.8 = Yes. Cutoff 0.6: Score 0.4 = No, Score 0.6 = Yes, Score 0.8 = Yes.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / AI-7/17 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Only 0.8 meets the new boundary.

Changed condition

Set the cutoff to exactly 0.6.

Worked changed-case result

0.6 and 0.8 are positive because the rule includes equality.
Why: The declared at-least boundary includes the threshold value.

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only 0.8 meets the new boundary. Changed condition: Set the cutoff to exactly 0.6. Expected result: 0.6 and 0.8 are positive because the rule includes equality.

Reason: The declared at-least boundary includes the threshold value.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

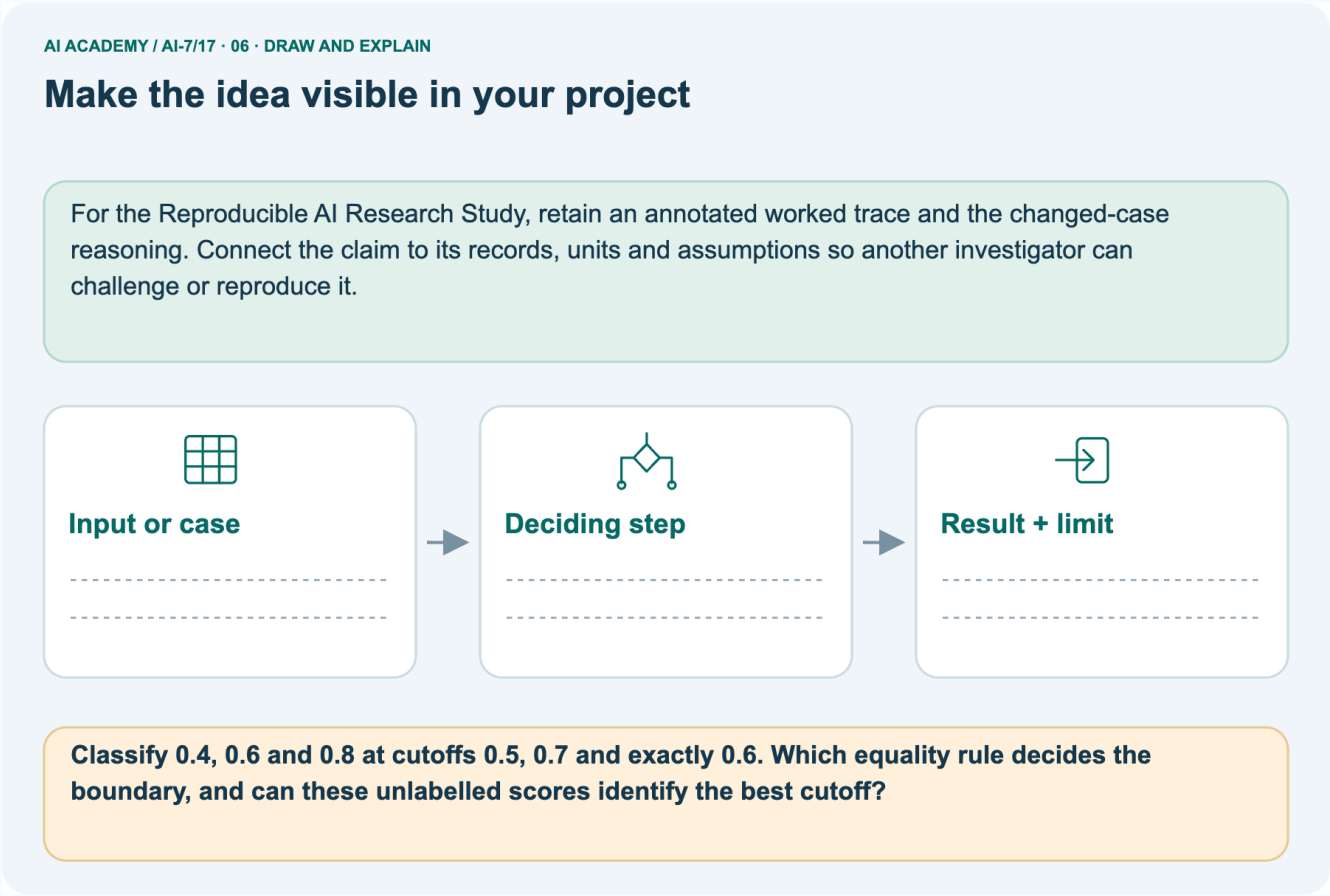
Sensitivity analysis checks how conclusions change under plausible choices. Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

Limit: Without reference labels, these counts do not identify the best cutoff or establish correctness.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ **Earlier wording, alternatives and picture transcripts**

01. Sensitivity analysis: the case

02. Follow the deciding steps

03. Score decisions under inclusive cutoffs

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.